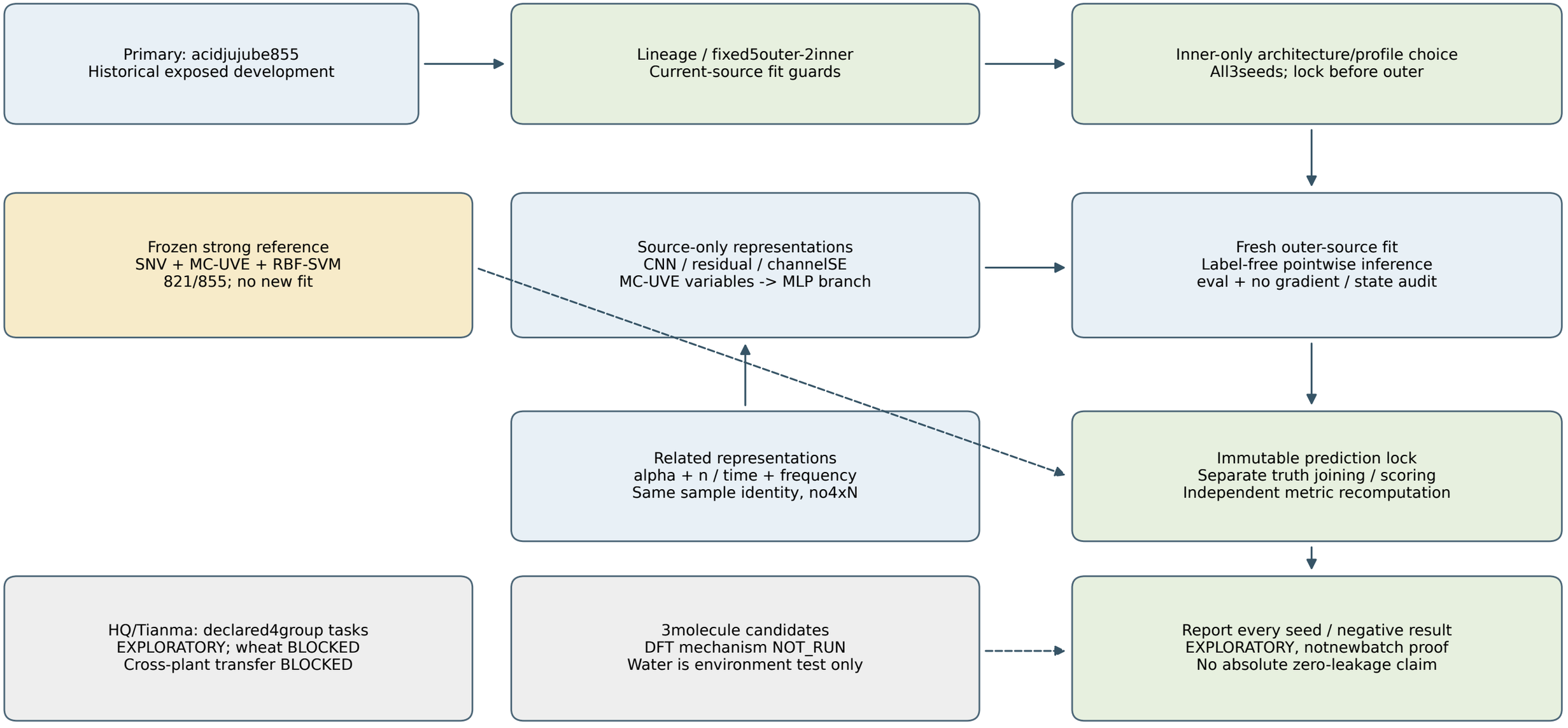Research4 upgrade: preserve acidjujube gains, add guarded neural comparison and bounded auxiliary work